SIMATS ENGINEERING
ASSESSMENT TOOL 3:

MCQ Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1596

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* (BL3)

Dave's Taxonomy: Precision (Level 3)
Question Count: 15
Duration : 60 Minutes

Total Marks: 30

Code of Conduct

I G. Amrutha (192511250) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Amrutha

MCQ Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Understanding	30%	Answers reflect strong and accurate concept mastery	Mostly accurate understanding	Partial understanding	Weak understanding
Accuracy of Responses	30%	Nearly all responses are correct	Most responses are correct	Some responses are correct	Few responses are correct
Application Awareness	15%	Shows strong recognition of practical cloud use cases	Reasonable recognition	Limited recognition	Weak recognition
Time Management	10%	Completes assessment efficiently	Completes on time	Needs extra time	Unable to finish well
Question Interpretation	15%	Interprets all items correctly	Misreads very few items	Misreads several items	Frequently misinterprets questions

Total Marks Awarded:

G. Amrutha
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MCQ TEST

CO3-AT3

1) A Company's cloud storage system requires 5 TB of data replication across three regions. What is the total storage required?

(a) 5TB (B) 10TB (C) 15TB (D) 20TB

A. (C) 15TB

2) A cloud storage provider offers a 99.99% uptime SLA. How much downtime can be expected per month?

(A) ~4 minutes (B) ~40 minutes

(C) ~4 hours (d) ~24 hours.

A. (A) ~4 minutes

3. A cloud-based load balancer distributes 60,000 requests per second among 10 servers. What is average number of requests per server per second. ?

(a) 5000 (b) 6000 (c) 7000 (d) 10,000

A. (C) 7000

4) A company's cloud network has 1Gbps bandwidth and needs to transfer 500GB of data. How long will it take?

(a) 1 hour (b) 10hours (c) 1.1hours (d) 5.5hours

5) A cloud storage system experiences a 99.95% availability rate. How much downtime does it experience per year?

(a) 01-4hours (b) ~8 hours

(c) ~20hours (d) ~40 hours.

A. (a) 01-4hours

6. A cloud provider offers pay-as-you-go pricing at $0.05 per GB for storage. If a company stores 10TB, what is the monthly cost?

(a) $50 (b) $500 (c) $1000 (d) $2000

A. (c) $1000

7. A cloud network has 10Gbps throughput but experiences 2% packet loss. what is the effective throughput?

(a) 9.8Gbps (b) 9.6Gbps

(c) 8.0Gbps (d) 7.5Gbps

A: (a) 9.8Gbps

8. A user uploads 250MB of data per day to the cloud. How much data will be uploaded in one year (365 days)?

(a) 91.25 GB (b) 85 GB

(c) 100 GB (d) 120 GB.

9. A cloud-based video streaming service requires 5 Mbps bandwidth per user. If 100 users are streaming simultaneously, what is the total required bandwidth?

(a) 500 Mbps (b) 250 Mbps

(c) 200 Mbps (d) 1000 Mbps.

A. (a) 500 Mbps

10. A cloud provider guarantees 99.95% uptime per year. How much downtime does this allow in minutes per year?

(a) 262 minutes (b) 365 minutes (c) 525 mins (d) 150 mins

A. (a) 262 minutes

11. A virtualized data center has 200 physical servers, If each server runs 40 virtual machines, what is the total number of VMs in the data centre.

(a) 4,000 (b) 8,000 (c) 6,000 (d) 10,000.

A. (b) 8,000

12. A physical server has 128 GB of RAM and hosts 10 virtual machines, each requiring 12 GB. How much memory is overcommitted?

(a) 4 GB (b) 6 GB (c) 8 GB (d) 10 GB.

A. (c) 8 GB

13. A virtual machine requires 4 vCPUs. If a server has a total of 64 cores and an over commitment ratio of 2:1, how many VMs can be hosted?.

(a) 16 (b) 32 (c) 64 (d) 128.

A. (b) 32

14. A company migrates 50 on-premise servers to the cloud, reducing power consumption by 60%. If the original power usage was 100 kW, what is the new power consumption?

(a) 40 kW (b) 50 kW (c) 60 kW (d) 70 kW.

A (a) 40 kW

15 A cloud provider operates a virtualized data center with 100 physical servers, each with 64 vCPUs. If the overcommitment ratio is 4:1, how many total vCPUs can be allocated?

(a) 6,400 (b) 12,800. (c) 25,600 (d) 50,000

A. (c) 25,600

16. which cloud deployment model is used for organizations that require both private and public cloud resources?

(A): Hybrid cloud (B): public cloud

(C): private cloud (D): community cloud.

A. (A): Hybrid cloud

17. which cloud security strategy ensures that sensitive data remains private even if intercepted?

(A): Encryption (B): Load balancing

(C): Virtualization (d): Resource pooling.

A (A): Encryption

18. If a hypervisor hosts 50 VMs and each VM uses 20GB bandwidth per day, Calculate total bandwidth consumption per week.

a) 5000 GB (b) 7000 GB (c) 6000 GB (d) 7500 GB

A. (b) 7000 GB

Q. A cloud system runs at 70% average CPU utilization. If it has 32 cores at 2.5 GHz each, what is total utilized processing power?

(a) 56GHz (b) 58GHz (c) 64GHz (d) 70GHz.

A. (a) 56GHz

20. If the network latency in a virtualized system is 5ms and processing time is 10 ms, calculate total round-trip delay for 100 requests.

(a) 1.5 sec (b) 2 sec (c) 1 sec (d) 3 sec.

A. (a) 1.5 sec